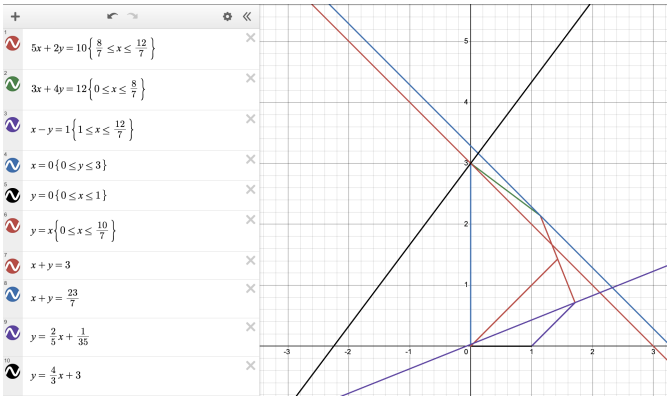
Homework 1 - Solutions

Problems 1, 2, and 3

Problem 1

(a) Feasible region. In the first quadrant, the feasible points lie below the lines $5x + 2y = 10$ and $3x + 4y = 12$, and above the line $y = x - 1$. The boundary segments in the Desmos graph enclose a feasible polygon with vertices in boundary order

$$(0,0), \quad (1,0), \quad \left(\frac{12}{7}, \frac{5}{7}\right), \quad \left(\frac{8}{7}, \frac{15}{7}\right), \quad (0,3).$$



(b) Optimal solution and value. The objective lines $x + y = k$ move northeast as k increases. Their last contact with the feasible polygon is the intersection of $5x + 2y = 10$ and $3x + 4y = 12$. Solving these equations gives $7x = 8$, hence

$$(x^*, y^*) = \left(\frac{8}{7}, \frac{15}{7}\right), \qquad \max(x + y) = \frac{8}{7} + \frac{15}{7} = \boxed{\frac{23}{7}}.$$

This point also satisfies $x - y = -1 \leq 1$ and $x, y \geq 0$, so it is feasible.

(c) Parameterized objective. The two edge directions from (x^*, y^*) are proportional to $(2, -5)$ and $(-4, 3)$. For (x^*, y^*) to remain optimal for $x + \lambda y$, neither edge direction may increase the objective:

$$(1, \lambda) \cdot (2, -5) = 2 - 5\lambda \leq 0, \qquad (1, \lambda) \cdot (-4, 3) = -4 + 3\lambda \leq 0.$$

Therefore

$$\boxed{\frac{2}{5} \leq \lambda \leq \frac{4}{3}}.$$

Because the feasible polygon is convex, checking the two adjacent edge directions is sufficient. At either endpoint of this interval, the adjacent edge also consists of optimal solutions.

Problem 2

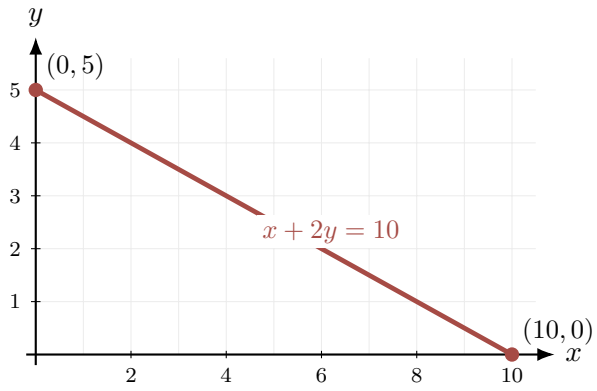
(a) Standard form. The equivalent maximization problem is

$$\begin{aligned} \text{maximize} \quad & y - x \\ \text{subject to} \quad & x + 2y \le 10, \\ & -x - 2y \le -10, \\ & x \ge 0, \quad y \ge 0. \end{aligned}$$

The first two inequalities together are equivalent to $x + 2y = 10$.

(b) Feasible region. The equation $x + 2y = 10$ has intercepts $(10, 0)$ and $(0, 5)$. Imposing $x, y \ge 0$ leaves exactly the closed segment joining these two points:

$$\left\{ (x, y) : y = \frac{10 - x}{2},\ 0 \le x \le 10 \right\}.$$



(c) Optimal solution and value. On the segment, $y = (10 - x)/2$, so

$$x - y = x - \frac{10 - x}{2} = \frac{3x - 10}{2}.$$

This expression increases with x on $0 \le x \le 10$. Thus its minimum occurs at $x = 0$, giving

$$\boxed{(x^*, y^*) = (0, 5)}, \qquad \boxed{\min(x - y) = -5}.$$

(d) Independent modifications. Each item below changes the original problem separately.

(i) Add $x + 2y = 5$. This contradicts the existing equality $x + 2y = 10$, so the LP is infeasible.

(ii) Remove $x \ge 0$. Setting $y = t \ge 0$ and $x = 10 - 2t$ gives feasible points with objective $x - y = 10 - 3t \to -\infty$ as $t \to \infty$. Hence the minimization LP is unbounded below.

(iii) Add $y = 2x$. Combining it with $x + 2y = 10$ yields $5x = 10$, so $(2, 4)$ is the only feasible point.

(iv) Replace the objective by $\max(x + 2y)$. Every point on the feasible segment satisfies $x + 2y = 10$. Therefore all points on that segment are optimal, with optimal value 10.

Problem 3

(a) Linear optimization model. Let a, b, c, d, e denote kilograms of apples, bananas, carrots, dates, and powdered egg whites, respectively. Set

$$x = \begin{pmatrix} a \\ b \\ c \\ d \\ e \end{pmatrix}, \qquad A = \begin{pmatrix} 0 & 0 & 8 & 0 & 0 \\ 46 & 123 & 60 & 4 & 0 \\ 60 & 50 & 330 & 390 & 1040 \\ 24 & 46 & 28 & 80 & 0 \\ 3 & 7 & 9 & 25 & 800 \end{pmatrix}, \qquad n = \begin{pmatrix} 1 \\ 30 \\ 400 \\ 15 \\ 15 \end{pmatrix}, \qquad p = \begin{pmatrix} 4 \\ 2 \\ 3 \\ 10 \\ 70 \end{pmatrix}.$$

The entries of Ax are the totals of vitamin A, vitamin C, calcium, fiber, and protein, in that order. The first three totals and requirements are measured in mg; the last two are measured in g. The price vector p is measured in dollars per kg. The required LP is

$$\boxed{\begin{array}{ll} \text{minimize} & p^T x = 4a + 2b + 3c + 10d + 70e \\ \text{subject to} & Ax \geq n, \\ & x \geq 0. \end{array}}$$

The inequality $Ax \geq n$ is componentwise, giving the five explicit requirements

$$\begin{aligned} 8c &\geq 1, \\ 46a + 123b + 60c + 4d &\geq 30, \\ 60a + 50b + 330c + 390d + 1040e &\geq 400, \\ 24a + 46b + 28c + 80d &\geq 15, \\ 3a + 7b + 9c + 25d + 800e &\geq 15. \end{aligned}$$

(b) Feasibility. The carrot column of A has a positive coefficient in every nutrient row. Because there is no upper bound on the amount of carrots, choosing only carrots with amount

$$c \geq \max\left\{\frac{1}{8}, \frac{30}{60}, \frac{400}{330}, \frac{15}{28}, \frac{15}{9}\right\} = \frac{5}{3}$$

meets all five requirements. In particular, for $x = (0, 0, 5/3, 0, 0)^T \geq 0$,

$$Ax = \begin{pmatrix} 40/3 \\ 100 \\ 550 \\ 140/3 \\ 15 \end{pmatrix} \geq \begin{pmatrix} 1 \\ 30 \\ 400 \\ 15 \\ 15 \end{pmatrix} = n.$$

Therefore the feasible set is nonempty, and the LP is feasible.

(c)

(d)